AARSH DESAI

AI Product Engineer

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SUMMARY

AI Product Engineer with hands-on experience across the deep-learning lifecycle, from model training and evaluation to production deployment. Built a multilingual voice AI and RAG pipeline for clinical documentation at Raaz MD, plus RAG assistants and LangGraph and tool-calling agents measured on labelled evaluation sets. Skilled in PyTorch, LLMs/RAG, and responsible ML practices, including safety guardrails and human-in-the-loop review.

EDUCATION

Purdue University, College of Science **Aug 2022 - May 2026**
Bachelor of Science, Data Science (Minor in Economics) *West Lafayette, IN*

WORK EXPERIENCE

Raaz MD **Jul 2026 - Sep 2026**
AI Product Engineer Intern

- Built an end-to-end multilingual voice-AI pipeline that turned Hindi and other regional-language calls into structured clinical summaries with RAG (ChromaDB) and a locally hosted 4-bit quantized Mistral-7B model, cutting manual review time by ~40%.
- Developed an automated call-type routing system that separated clinical follow-ups from logistics escalations, reducing manual review time.
- Implemented reliability safeguards, adding prompt-level plausibility guardrails to catch LLM hallucinations and a PII de-identification layer (NER-based redaction and phone-number pseudonymization), which lowered false risk assessments and ensured compliant handling of sensitive data.

Purdue University **Aug 2025 - Dec 2025**
Undergraduate Teaching Assistant

- Enabled 50+ first-year students to complete coding assignments independently by teaching Python, Git, and command-line workflows and breaking down technical concepts in weekly sessions.

Oeson Learning **May 2024 - Aug 2024**
Data Analyst Intern

- Built automated Python pipelines to ingest and validate 50,000+ data rows, improving baseline reliability by 20%, and designed a pipeline-testing framework with continuous A/B tracking, driving a 9% lift in system interaction metrics.
- Cut stakeholder reporting time 40% by building Tableau dashboards that surfaced model and KPI outputs for technical and non-technical teams.

PROJECTS

CardioDesk: Patient-Message Triage Assistant https://github.com/aarshdesai-ds/cardiodesk-patient-message-assistant

- Built and deployed a RAG assistant that triages patient messages for nurse review; its two-layer safety check caught 100% of emergencies (rules alone 69%, model alone 81%), with 98% intent accuracy.
- Rebuilt triage as a LangGraph graph with an interrupt() nurse-review step and FastAPI case endpoints, pausing 100% of normal drafts for review while 0% of emergencies waited and no results changed.

QuizMate: Python Tutor with Long-Term Memory https://github.com/aarshdesai-ds/quizmate-python-tutor

- Built a LangGraph tutor with a quiz subgraph, interrupt() pauses, and cross-session SQLite memory; found LLM-written quizzes put 64% of right answers on one letter, and cut that to 31% by shuffling in code.
- Designed long-term memory with fact extraction, code-only score writes, and approval before deletes; traced each eval failure to a mislabel or a rule clash, raising extraction accuracy from 90% to 100%.

FitBuddy: Home-Workout Chatbot with an Injury Guard https://github.com/aarshdesai-ds/fitbuddy-workout-chatbot

- Built a LangGraph workout chatbot with four tools, Chroma retrieval, SQLite-checkpointed memory, and LangSmith tracing; an injury guard cut the unsafe-reply rate from 50% to 0% on labelled test questions.
- Added human-in-the-loop approval with interrupt() before logging, and rewrote a keyword safety check that had missed 78% of unsafe replies, validating the new rule with 19 hand-written test cases.

Structural Heart Disease Detection from 12-Lead ECG https://github.com/aarshdesai-ds/echonext-shd-detection

- Built a 1D-CNN residual ensemble with ECG and clinical feature fusion on 82,543 ECGs, reaching 0.842 AUROC and 0.812 AUPRC with calibration and ablations.
- Containerized the end-to-end AI application using FastAPI and Docker, with a Gradio demo, continuous integration via GitHub Actions, and a versioned model registry.

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate: Aug 2024
- SQL for Data Science, University of California, Davis: Oct 2024
- IBM AI Engineering Professional Certificate and IBM Data Engineering Professional Certificate: In Progress

TECHNICAL SKILLS

- **AI Engineering:** LLMs, RAG, LangGraph, LangChain, LangSmith, Tool-Calling Agents, Human-in-the-Loop Approval, Long-Term Memory, LLM Evaluation, Structured Output, Prompt Engineering, Hallucination Guardrails, OpenAI and Gemini APIs, ChromaDB, Fine-Tuning (LoRA), Hugging Face
- **Machine Learning & Deep Learning:** Python, PyTorch, Scikit-learn, TensorFlow/Keras, Pandas, NumPy, CNNs, Transfer Learning, Random Forest, XGBoost
- **Deployment, MLOps & Evaluation:** FastAPI, Docker, AWS (EC2, S3), GitHub Actions CI, Model Registry, Git/GitHub, Gradio, Streamlit, Calibration, AUROC/AUPRC